Department of Research
&
Graduate Studies

EMA 5104: Methods of Teaching Mathematics at the Secondary Level

Lecture Notes: 1

Lecturer: Peter Wintz

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Common Mistakes Teachers Make

- Belittling
- Humiliating
- Criticising
- Fault-finding
- Threatening

Common Mistakes Teachers Make

- Mortification
- Neglect/Disregard
- Pampering
- Instilling fear & anxiety
- Nagging
- Physical punishment

Children Respond To:

- Consistency
- Flexibility
- Fairness
- Natural confidence
- Love and affection
- Respect
- Attention

What is Mathematics?

Mathematics has no complete definition, it has various interpretations and various basic ideas depending on how one looks at it. The confusion in the nature of mathematics was evident in 1899 when Frege in corresponding to Hilbert (in Scott-Hodgetts, et al, 1990, p. 200), stated that:

> There is a widespread confusion with regard to definitions of mathematics ... it seems to me that complete anarchy and subjective caprice now prevail.

What is Mathematics?

- Mathematics is a set of rules and procedures.
- Mathematics is a tool for everyday life.
- Mathematics is an application of logical thinking and/or step-by-step method.
- Mathematics is experimental and not a static body of knowledge.

Dougherty (1990, p. 123)

Mason (1988) points out that mathematics is associated (to a great extent) with seeing generalities which take place internally, in the imagination. This brings us to the third point. Mathematics is an application of logical thinking and/or step-by-step methods. The process of logical thinking can only take place in the mind and is hinged on the ability to 'figure out' that particular situation. Schoenfeld (1988, p. 8) seems to share this view when he writes indeed, 'figuring it out' is what mathematics is all about".

What is Mathematics?

In sum, mathematics can be seen as a subject which:

- is composed of unique symbols and is guided by sets of rules and procedures
- can be used to inform most decision making processes (such as budgeting and construction) and facilitates other school subjects requires some level of imagination, reasoning and critical thinking processes.

The first case listed is insufficient. Mathematics must be more than a set of rules and procedures. Here, apparently, the doing of mathematics is perceived wholly as the writing down of formulas and/or other symbolic operations rather than the reasoning and thinking process involved. The limited perception of mathematics as a set of rules, according to Davis (1986), is the 'wrong view of mathematics'.

The fourth case - mathematics is experimental and not a static body of knowledge - relates to Cockcroft's (1982, para 249) observations, "Mathematics is only 'useful' to the extent which it can be applied to particular situations and is the ability to apply mathematics to a variety of situations". 'A static body of knowledge' is limited if this body of knowledge is to be applied to a variety of situations since the usefulness of mathematics may only be realised if one experiments with it. Consequently, "[mathematics] may enrich our understanding, enabling us to relate and interpret our experience as well as giving pleasure" (Bell, et al, 1978, p. 2).

What does it mean to learn Mathematics?

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... learning, or coming to know, is an adaptive process which tries to make sense of experience ... knowledge for any learner, can be more than their own construction ...

(Jaworski, 1992, p. 12)

What does it mean to learn Mathematics?

Having examined some views on the nature of mathematics learning, the following recommendations are presented.

- Proper learning, or coming to know, is unlikely if teachers depend on transmitting knowledge. Knowledge is not a commodity which can be communicated (von Glaserfeld, 1990).
- Mathematical knowledge is acquired only when the learner, actively builds up (constructs) his/her own knowledge. Nonetheless, there will always be the possibility of common mathematical knowledge among learners [Edwards and Mercer (in Jaworski, 1992, p. 13)].

END